

A gamified introduction to Python Programming

Lecture 8 The list type and its uses

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Outline (Chapter 8)

- What is the **list type**, in detail, this time?
- What are the **indexing, slicing, updating** and **traversing** a list operations?
- What are **lists of lists** and their uses?
- How to **create, add** and **remove** elements from a list?
- What are advanced operations on list such as **concatenation, repetition, insertion**, etc?
- What does **RTFM** mean?
- **Practice** on lists.

The list type

Definition (lists): a **list** is a **sequence** of several variables or elements, listed, in order, between **brackets** and separated by **commas**.

Lists can contain variables of any types (int, float, string, etc.) and lists can also contain mixed types of variables.

A list with no elements, is called an **empty list** and is noted **[]**.

```
1 a_list = [0, 1, 2, 3, 4]
2 print(a_list)
3 print(type(a_list))
```

```
[0, 1, 2, 3, 4]
<class 'list'>
```

```
1 another_list = ['a', 'b', 'c', 'd']
2 print(another_list)
```

```
['a', 'b', 'c', 'd']
```

```
1 a_float = 3.14
2 an_int = 10
3 a_string = 'Hello'
4 mixed_list = [a_float, an_int, a_string]
5 print(mixed_list)
```

```
[3.14, 10, 'Hello']
```

Length of a list

Definition (**length** of a list):

The **length** of a list consists of the **number of elements in the list**.

- It can be simply computed by using the **len()** function on a list, which returns a **positive integer** corresponding to the **number of elements** in the list.
- An **empty list**, defined as [], has zero elements in it.

```
1 a_list = [0, 1, 2, 3, 4]
2 print(len(a_list))
```

5

```
1 another_list = ['a', 'b', 'c', 'd']
2 print(len(another_list))
```

4

```
1 an_empty_list = []
2 print(an_empty_list)
3 print(len(an_empty_list))
```

[]

0

Indexing/Accessing the elements of a list

Indexing/Accessing: The elements in a list are **indexed**, with their positions identified as integers. They can be accessed using the `[]` notation, on a list, with an integer **index** value in between square brackets.

- The **first element** of the list has the **index 0** (remember, we start counting from 0 in Python!).
- the **second element** has index **1**, etc.
- The last element has index **$n - 1$** (not **$n!$**), with **n** being the **length** of the list.

```
1 my_list = ['a', 'b', 'c', 'd']  
2 print(my_list)
```

```
['a', 'b', 'c', 'd']
```

```
1 print(my_list[0])
```

```
a
```

```
1 print(my_list[1])
```

```
b
```

```
1 print(my_list[3])
```

```
d
```

```
1 print(my_list[4])
```

```
-----  
IndexError                                Tra  
<ipython-input-11-df42a28462c1> in <module>  
----> 1 print(my_list[4])
```

```
IndexError: list index out of range
```

Indexing/Accessing the elements of a list

When indexing/accessing elements in a list, we can also use **negative positional indexes**.

- **-1** refers to the **last element** of the list,
- **-2** the **second to last element**, etc.
- And **-*n*** is therefore the index of the **first element**.

Each element can therefore be identified with two positional indexes (one positive, one negative).

```
1 my_list = ['a', 'b', 'c', 'd']  
2 print(my_list)
```

```
['a', 'b', 'c', 'd']
```

```
1 print(my_list[-1])
```

```
d
```

```
1 print(my_list[-2])
```

```
c
```

```
1 print(my_list[-4])
```

```
a
```

```
1 print(my_list[-5])
```

```
-----  
IndexError                                Tra  
Cell In[10], line 1  
----> 1 print(my_list[-5])  
  
IndexError: list index out of range
```

Slicing a list

Definition (**slicing**):

Slicing is used to retrieve a **subset** of the list, by specifying to indexes, separated with a colon (:) symbol. **Slicing** with **[a:b]** returns another list, containing all elements of the original list:

- Starting **from positional index a** (with a included),
- All the way **to positional index b** (with b **not** included).

```
1 my_list = [1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
2 print(my_list)
```

```
[1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
```

```
1 print(my_list[0:4])
```

```
[1, 4, 9, 14]
```

```
1 print(my_list[6:8])
```

```
[27, 38]
```

```
1 print(my_list[-7:-3])
```

```
[14, 15, 16, 27]
```

```
1 print(my_list[4:-2])
```

```
[15, 16, 27, 38]
```

Note: Intuitively, it seems to have a roughly similar behavior to the **range()** function!

Slicing a list

Additional rule: Omitting a value in a slicing, either a or b, has effects.

- **[:b]** means:
 - all elements from the beginning of the list,
 - until index b (with b not included).
- **[a:]** means:
 - all element from index a,
 - until the end of the list (last element included).

```
1 my_list = [1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
2 print(my_list)
```

```
[1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
```

```
1 print(my_list[:5])
```

```
[1, 4, 9, 14, 15]
```

```
1 print(my_list[7:])
```

```
[38, 49, 50]
```


Slicing a list

Additional rule #2: Slicing with three values **[a:b:c]** returns another list as a result, containing all elements of the original list:

- Starting **from index a** (with element in position a included),
- All the way **to index b** (with element in position b **not** included),
- With **steps of size c**.

Note: Behavior similar to `range()`.

```
1 my_list = [1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
2 print(my_list)
```

```
[1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
```

```
1 print(my_list[:5])
```

```
[1, 4, 9, 14, 15]
```

```
1 print(my_list[7:])
```

```
[38, 49, 50]
```

```
1 print(my_list[0:6:2])
```

```
[1, 9, 15]
```

```
1 print(my_list[-1:-7:-3])
```

```
[50, 27]
```



Which one of the statements below allows to display all the elements of a list `my_list`, in reverse order?

① Start presenting to display the poll results on this slide.

Slicing a list

```
1 # Original list
2 my_list = [1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
3 print(my_list)
4 # A
5 print(my_list[-1:(-len(my_list))])
6 # B
7 print(my_list[-len(my_list):-1])
8 # C
9 print(my_list[-1:(-len(my_list)):-1])
10 # D
11 print(my_list[-1:(-len(my_list) - 1):-1])
```

```
[1, 4, 9, 14, 15, 16, 27, 38, 49, 50]
[]
[1, 4, 9, 14, 15, 16, 27, 38, 49]
[50, 49, 38, 27, 16, 15, 14, 9, 4]
[50, 49, 38, 27, 16, 15, 14, 9, 4, 1]
```

```
1 print(my_list[::-1])
```

```
[50, 49, 38, 27, 16, 15, 14, 9, 4, 1]
```

Activity 1 - Distributing a deck of cards

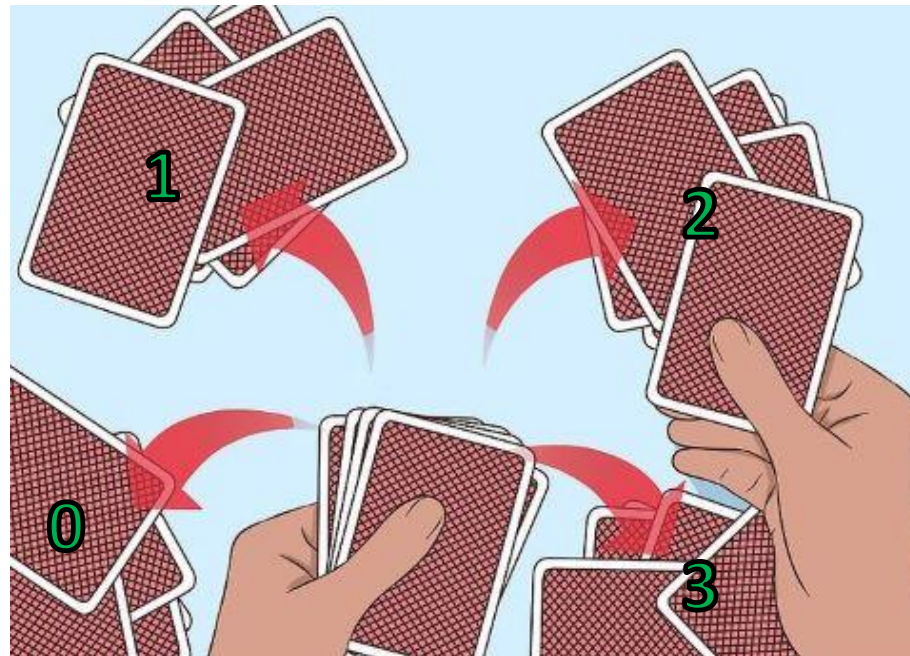
- Let us consider a standard, shuffled deck of cards, defined as a list.

```
cards_deck = ['3 of Spades', 'Jack of Diamonds', '2 of Hearts', '4 of Clubs', '8 of Clubs', \
              '4 of Spades', '9 of Hearts', 'Jack of Spades', '2 of Diamonds', '9 of Spades', \
              'Ace of Hearts', 'Jack of Hearts', '4 of Hearts', 'Queen of Clubs', '9 of Clubs', \
              'King of Hearts', 'Ace of Clubs', 'Ace of Diamonds', '8 of Diamonds', 'King of Diamonds', \
              '6 of Diamonds', '7 of Spades', 'King of Clubs', '10 of Clubs', '9 of Diamonds', \
              '3 of Hearts', '7 of Diamonds', '6 of Clubs', '4 of Diamonds', '3 of Diamonds', \
              '6 of Spades', '5 of Hearts', 'Jack of Clubs', 'King of Spades', '5 of Diamonds', \
              '10 of Spades', 'Queen of Hearts', '2 of Clubs', '2 of Spades', 'Ace of Spades', \
              'Queen of Spades', '5 of Clubs', '7 of Clubs', '6 of Hearts', '10 of Diamonds', \
              '10 of Hearts', 'Queen of Diamonds', '5 of Spades', '3 of Clubs', '7 of Hearts', \
              '8 of Hearts', '8 of Spades']
```

- In this activity, we will distribute the cards between several players.

Activity 1 - Distributing a deck of cards

- Let us now consider that we have **n players**.
 - The players are referred to as **player 0, player 1, ..., player (n-1)**.
 - Players are seating around a table in a **clockwise** manner: player 1 is seating on the left of player 0, player 2 is seating on the left of player 1, etc.



Activity 1 - Distributing a deck of cards

Write a function **distribute_cards()**, which

- **receives** a shuffled deck of cards, **cards_deck**, as its first input parameter,
- **receives** a number of players seating at the table, **players_number**, as its second input parameter,
- **receives** an integer **player_index**, with value in $[0, \text{players_number} - 1]$, as its third input parameter. This corresponds to a specific player sitting at the table.
- **receives** an integer **start_from**, with value in $[0, \text{players_number} - 1]$, as its fourth input parameter. This value corresponds to the first player to receive a card.

Activity 1 - Distributing a deck of cards

The function **distribute_cards()** then

- **distributes all cards** in the deck, **one card at a time**, in **clockwise order**, between all players,
- **starting from the player** with index **start_from**.
- It is ok if some players have one more card than other players (for instance, if the number of cards is not divisible by number of players)

The function **distribute_cards()** returns a list of cards, **player_hand**, corresponding to the **hand** of player with index **player_index**, after we are done distributing the cards. Find a smart slicing for the deck!

Traversing lists (element-wise)

You have seen how we can give lists to **for** statements to traverse and retrieve each element **one-by-one**.

1. The “**for element in list:**” approach is commonly referred to as traversing a list **element-wise**.

```
1 my_list = [1, 4, 9, 14, 15]
2 print(my_list)
```

```
[1, 4, 9, 14, 15]
```

```
1 # Element-wise
2 for element in my_list:
3     print("--")
4     print(element)
```

```
--
1
--
4
--
9
--
14
--
15
```


Traversing lists (index-wise)

You have seen how we can give lists to **for** statements to traverse and retrieve each element **one-by-one**.

2. The “**for index in range(len(list)):**” approach is commonly referred to as traversing a list **index-wise**.

```
1  # Index-wise
2  for index in range(len(my_list)):
3      print("--")
4      print(index, my_list[index])
```

```
--
0  1
--
1  4
--
2  9
--
3  14
--
4  15
```

Traversing lists (index and element-wise)

You have seen how we can give lists to **for** statements to traverse and retrieve each element **one-by-one**.

3. You have also seen how to use the **enumerate()** generator, and how it allows to retrieve **both the indexes and elements** at the same time, while traversing a list.

```
1  # Index and element-wise
2  for index, element in enumerate(my_list):
3      print("--")
4      print(index, element)
```

```
--
0 1
--
1 4
--
2 9
--
3 14
--
4 15
```

Traversing lists (while)

You have seen how we can give lists to **for** statements to traverse and retrieve each element **one-by-one**.

4. You can also replace the **for** loop with a **while** loop.
(Not recommended however!)

```
1 index = 0
2 while index < len(my_list):
3     print("--")
4     print(index, my_list[index])
5     index += 1
```

```
--
0 1
--
1 4
--
2 9
--
3 14
--
4 15
```

Updating an element in a list

Definition (**Updating a list**):

The indexing notation

list[index] = value can be used to **update** an element in the list.

- The accessed element in given position index (with the = sign symbol) receives a new value. This replaces the current value in the list with a new value.
- **Note:** An element can be updated with a completely different type. Also works with slicings.

```
1 my_list = [1, 4, 9, 14, 15]
2 print(my_list)
```

```
[1, 4, 9, 14, 15]
```

```
1 my_list[2] = 'Hello'
2 print(my_list)
```

```
[1, 4, 'Hello', 14, 15]
```

```
1 my_list[1:4] = 'Queen', 'of', 'Hearts'
2 print(my_list)
```

```
[1, 'Queen', 'of', 'Hearts', 15]
```

```
1 another_list = [7, 1, 9]
2 my_list[1] = another_list
3 print(my_list)
```

```
[1, [7, 1, 9], 'of', 'Hearts', 15]
```

Lists of lists

Observation: Elements of a list can be any type, therefore an element of a list can be another list! This is called a **list of lists**.

- Careful with the **len()** function! If an element of a list is a list itself, it still only counts as 1 element for the main list!
- You can **use multiple indexing**, i.e. using **[]** several times in a row, to **access the elements of a sublist**.
- Very useful to represent **tables** (as in Excel) or **matrices** (something for later).

```
1 my_list = [1, [7, 1, 9], 14, 15]
2 print(my_list)
```

```
[1, [7, 1, 9], 14, 15]
```

```
1 print(len(my_list))
```

```
4
```

```
1 print(my_list[0])
```

```
1
```

```
1 print(my_list[1])
```

```
[7, 1, 9]
```

```
1 print(len(my_list[1]))
```

```
3
```

```
1 print(my_list[1][0])
```

```
7
```

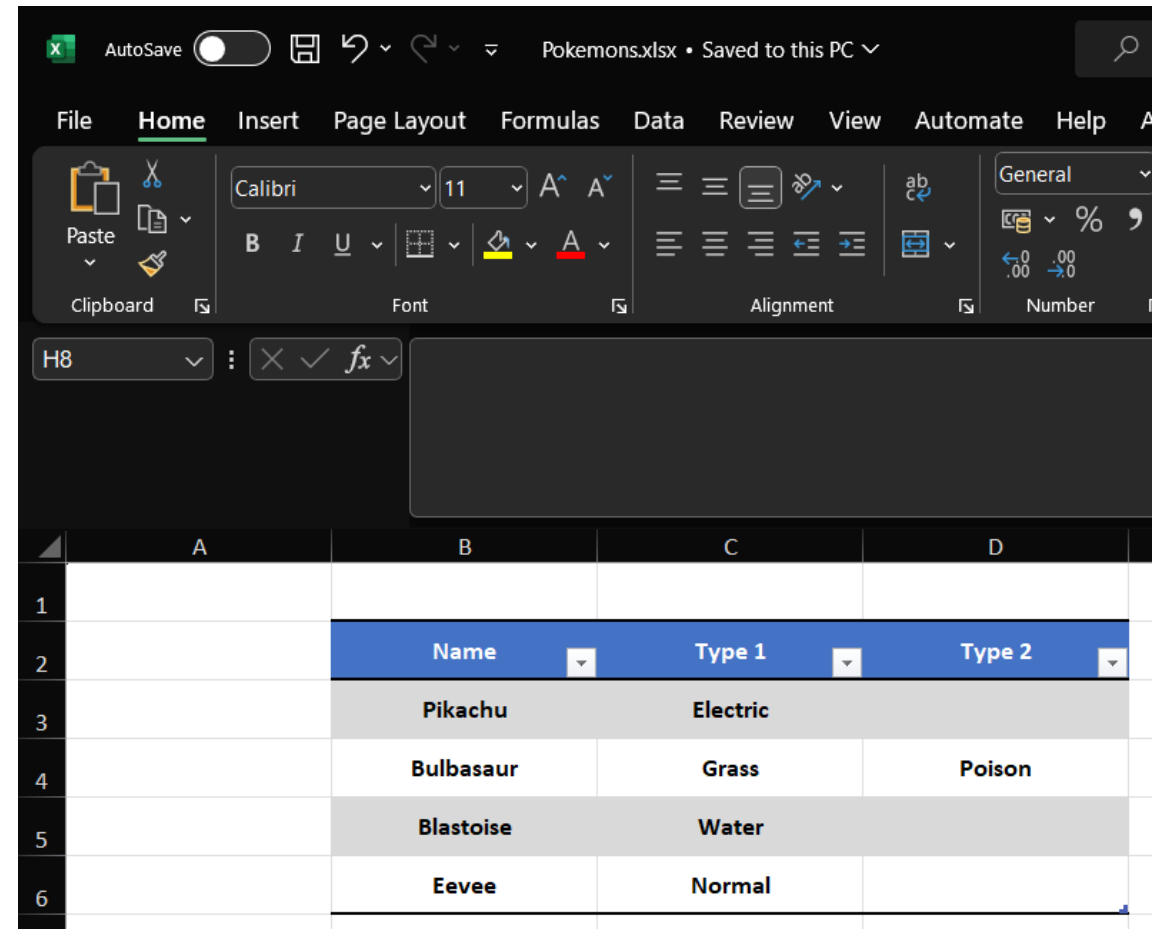
```
1 print(my_list[1][2])
```

```
9
```

Lists of lists

Lists of lists are commonly used in Python to represent **tables of values**.

Widely used in **Data Science** because **Python** allows for more processing operations than **Excel**!



The screenshot shows the Microsoft Excel interface with a file named 'Pokemons.xlsx'. The 'Home' tab is selected in the ribbon. The worksheet contains a table with 4 columns (A, B, C, D) and 6 rows. The first row (row 2) contains headers: 'Name', 'Type 1', and 'Type 2'. The subsequent rows (rows 3-6) contain data for different Pokemon: Pikachu (Electric), Bulbasaur (Grass, Poison), Blastoise (Water), and Eevee (Normal). The 'Type 2' column is empty for all entries.

	A	B	C	D
1				
2		Name	Type 1	Type 2
3		Pikachu	Electric	
4		Bulbasaur	Grass	Poison
5		Blastoise	Water	
6		Eevee	Normal	

```
my_list = [{"Name", "Type 1", "Type 2"}, # Columns labels  
           [{"Pikachu", "Electric", None}, # Entry 1  
            [{"Bulbasaur", "Grass", "Poison"}, # Entry 2  
             [{"Blastoise", "Water", None}, # Entry 3  
              [{"Eevee", "Normal", None}]] # Entry 4  
print(my_list)
```

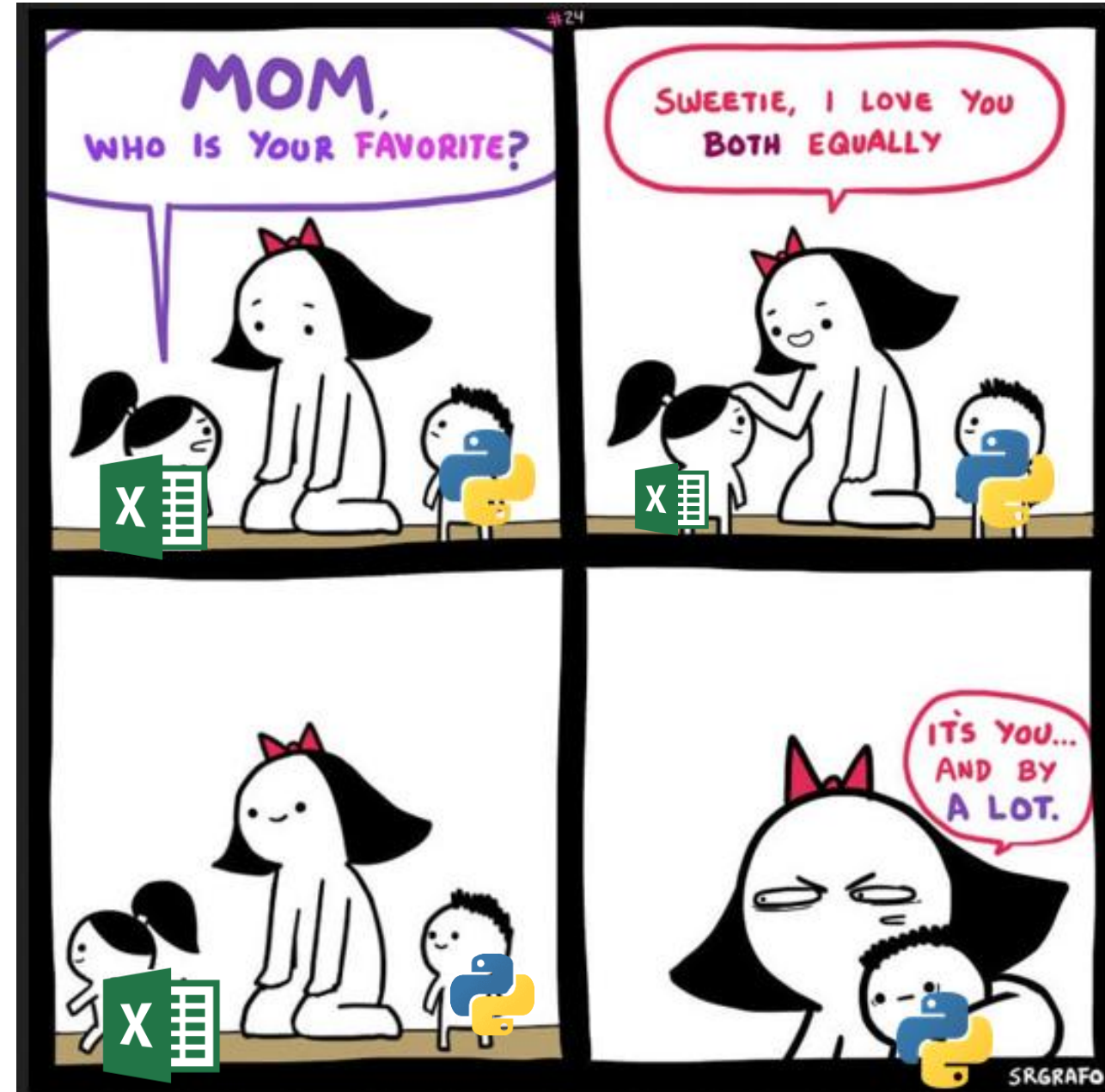
```
[['Name', 'Type 1', 'Type 2'], ['Pikachu', 'Electric', None], ['Bulbasaur', 'Grass', 'Poison'], ['Blastoise', 'Water', None], ['Eevee', 'Normal', None]]
```

Lists of lists

Lists of lists are commonly used in Python to represent **tables of values**.

Widely used in **Data Science** because **Python** allows for more processing operations than **Excel**!

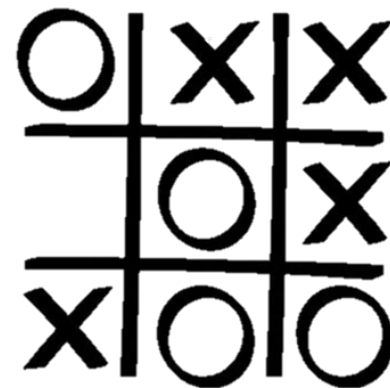
Great news: this means that you will no longer need Excel!



Activity 2 - TicTacToe

A **tic tac toe board** can be represented as a **list of lists**, by

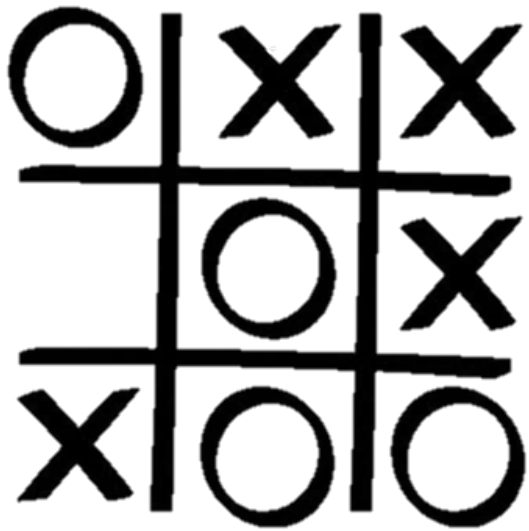
- defining a list of three elements,
 - each element is a sublist of three elements corresponding to each line, from up to down,
 - each sublist contains three elements, corresponding to each position on the line, from left to right,
- element will have a value **0** if the position is empty,
 - element will have a value **1** if the position contains a **circle**,
 - element will have a value **2** if the position contains a **cross**,



```
1 board = [[1, 2, 2], \
2           [0, 1, 2], \
3           [2, 1, 1]]
```


Activity 2 - TicTacToe

A tic tac toe board can be represented as a list of lists.



Write a function **board_winner()**, which:

- receives a board, defined as a list of lists, as its sole input parameter,
- returns a single output, which consists of the value
 - 1 if the circles won,
 - 2 if the crosses won,
 - and 0 if there is no clear winner.

```
1 board = [[1, 2, 2], \
2           [0, 1, 2], \
3           [2, 1, 1]]
```

Lists of lists

Traversing lists of lists requires the use of **nested for** loops.

The **first for loop** browses through the **lines**, which are the **sublists of the main list**, one line at a time.



```
2 for line in my_list:
3     print("-----")
4     print("Line: ", line)
5     for element in line:
6         print("-")
7         print(element)
```

```
-----
Line:  ['Name', 'Type 1', 'Type 2']
-
Name
-
Type 1
-
Type 2
-----
Line:  ['Pikachu', 'Electric', None]
-
Pikachu
-
Electric
-
None
-----
Line:  ['Bulbazaaur', 'Plant', 'Poison']
```

Lists of lists

Traversing lists of lists requires the use of **nested for** loops.

The **first for loop** browses through the **lines**, which are the **sublists of the main list**, one line at a time.

The **second for loop** browses through the **elements in the sublists**, one element at a time.

```
1  for line in my_list:
2      print("-----")
3      print("Line: ", line)
4      for element in line:
5          print("-")
6          print(element)
```



```
-----
Line:  ['Name', 'Type 1', 'Type 2']
-
Name
-
Type 1
-
Type 2
-----
Line:  ['Pikachu', 'Electric', None]
-
Pikachu
-
Electric
-
None
-----
Line:  ['Bulbazzard', 'Plant', 'Poison']
```

Membership: the **in** keyword

We have seen: the **in** keyword can be used to traverse lists in for loops (it connects iteration variables and lists/generators).

New!: The **in** keyword can also be used as a Boolean question for membership (is element in list?).

- If the element appears in the list, then the result of the operation is a Boolean **True**.
- And **False** otherwise.

```
1 my_list = [1, 7, 10, 14, 15]
2 print(my_list)
```

[1, 7, 10, 14, 15]

```
1 for element in my_list:
2     print("--")
3     print(element)
```

--
1
--
7
--
10
--
14
--
15



```
1 bool1 = (7 in my_list)
2 print(bool1)
```

True



```
1 bool2 = (8 in my_list)
2 print(bool2)
```

False

Concatenation: + operator overload for lists

Concatenation: you can merge two lists together, by using the + operator.

- When summing two lists with +, Python does not sum the elements, as with numbers or vectors in mathematics.
- Instead, it behaves as with **strings** and concatenates both.

```
1 one_number = 6
2 another_number = 7
3 number_sum = one_number + another_number
4 print(number_sum)
```

13

```
1 a_string = '5'
2 another_string = '5'
3 string_sum = a_string + another_string
4 print(string_sum)
```

55

```
1 a_list = [0, 1, 2]
2 another_list = [1, 4, 7]
3 list_sum = a_list + another_list
4 print(list_sum)
```

[0, 1, 2, 1, 4, 7]

Adding an element with the `append()` method

Adding/Appending an element:

you can add an element at the end of an existing list with the `append()` method.

- The `append()` method modifies the list on which the method is applied.

- No need to reassign (That would be a mistake!). This would be incorrect

`a_list = a_list.append(an_element)`

```
1 a_list = [0, 1, 2]
2 an_element = 7
3 # Append method on a list
4 a_list.append(an_element)
5 print(a_list)
```

```
[0, 1, 2, 7]
```



What happens to `my_list` after
the operation
`my_list = my_list.append(7)`?

① Start presenting to display the poll results on this slide.

Activity 3 - Party merger

In many video games, you might form a group of friends and play together. In this activity, we will define a function, which attempts to merge two groups into a single one.

- A group will simply consist of a list of names, as shown below.

```
group = ["Matt", "Oka", "Tony",  
         "Norman"]
```

- Write a function **merge_groups()**, which receives two groups of people as lists of names.

- We will call these two groups variables, **group1** and **group2** respectively.
- This function will **merge both groups** into a new one containing the members of both groups, and will return the **merged_group** list as its only output.
- However, it will only do so, **if the resulting merged group does not contain more than 6 people**. Otherwise, the function will return both group1 and group2, therefore refusing to merge them.

Activity 4 – Complete, sorted deck of cards

A standard deck of cards consists of 52 different cards, forming all 52 possible combinations of 4 possible suits and 13 possible values ($4 \times 13 = 52$!)

- The 4 possible suits are listed in the **suits_list** below.

```
suits_list = ["Hearts", "Diamonds",  
"Spades", "Clubs"]
```

- The 13 possible values are listed in the **values_list** below.

```
values_list = ["Ace", "2", "3", "4",  
"5", "6", "7", "8", "9", "10", "Jack",  
"Queen", "King"]
```

Activity 4

– Complete, sorted deck of cards

Write a function **complete_deck()**,

- which receives **suits_list** and **values_list**, as input parameters,

suits_list = ["Hearts", "Diamonds", "Spades", "Clubs"]

values_list = ["Ace", "2", "3", "4", "5", "6", "7", "8", "9", "10", "Jack", "Queen", "King"]

- And returns a complete sorted deck, as defined on the right.

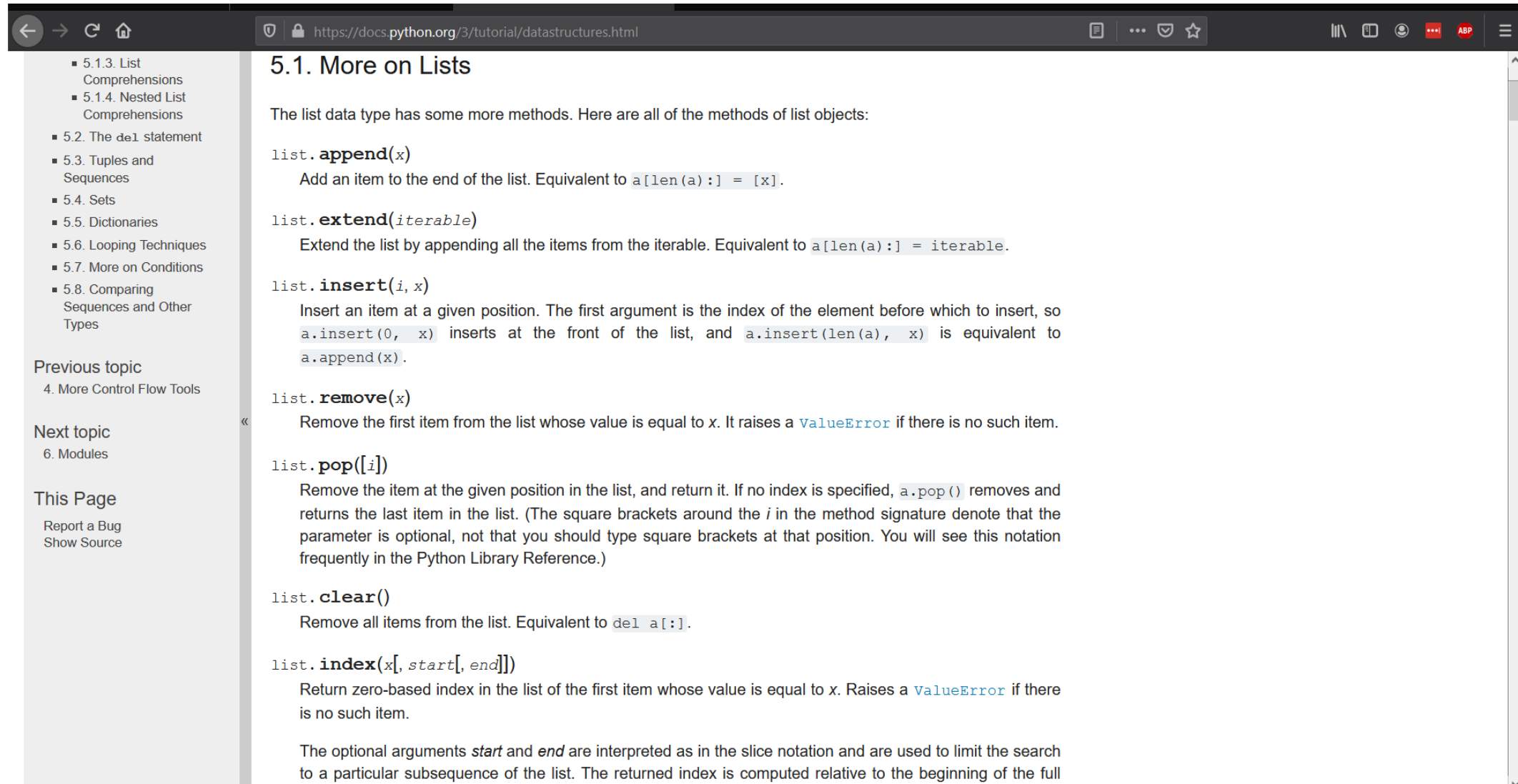
deck = ['Ace of Hearts', '2 of Hearts', '3 of Hearts', '4 of Hearts', ..., 'Queen of Hearts', 'King of Hearts', 'Ace of Diamonds', '2 of Diamonds', '3 of Diamonds', '4 of Diamonds', ..., 'Queen of Diamonds', 'King of Diamonds', 'Ace of Spades', '2 of Spades', '3 of Spades', '4 of Spades', ..., 'Queen of Spades', 'King of Spades', 'Ace of Clubs', '2 of Clubs', '3 of Clubs', '4 of Clubs', ... 'Queen of Clubs', 'King of Clubs']

More methods and functions on lists

The lists objects in Python have **many more built-in methods and functions** you could use.

- We **do not expect you to know** about all these methods and functions, nor about how they work in detail.
- Just know where to find the information if you need it.
- Also, do not be afraid to **read the Python documentation!**
- **List documentation:**
<https://docs.python.org/3/tutorial/datastructures.html>

More methods and functions on lists



The screenshot shows a web browser displaying the Python documentation for lists. The address bar shows the URL `https://docs.python.org/3/tutorial/datastructures.html`. The left sidebar contains a table of contents with links to various topics, including '5.1.3. List Comprehensions', '5.1.4. Nested List Comprehensions', '5.2. The `del` statement', '5.3. Tuples and Sequences', '5.4. Sets', '5.5. Dictionaries', '5.6. Looping Techniques', '5.7. More on Conditions', and '5.8. Comparing Sequences and Other Types'. Below the sidebar, there are links for 'Previous topic' (4. More Control Flow Tools), 'Next topic' (6. Modules), and 'This Page' (Report a Bug, Show Source). The main content area is titled '5.1. More on Lists' and contains the following text:

The list data type has some more methods. Here are all of the methods of list objects:

`list.append(x)`
Add an item to the end of the list. Equivalent to `a[len(a):] = [x]`.

`list.extend(iterable)`
Extend the list by appending all the items from the iterable. Equivalent to `a[len(a):] = iterable`.

`list.insert(i, x)`
Insert an item at a given position. The first argument is the index of the element before which to insert, so `a.insert(0, x)` inserts at the front of the list, and `a.insert(len(a), x)` is equivalent to `a.append(x)`.

`list.remove(x)`
Remove the first item from the list whose value is equal to `x`. It raises a `ValueError` if there is no such item.

`list.pop([i])`
Remove the item at the given position in the list, and return it. If no index is specified, `a.pop()` removes and returns the last item in the list. (The square brackets around the `i` in the method signature denote that the parameter is optional, not that you should type square brackets at that position. You will see this notation frequently in the Python Library Reference.)

`list.clear()`
Remove all items from the list. Equivalent to `del a[:]`.

`list.index(x[, start[, end]])`
Return zero-based index in the list of the first item whose value is equal to `x`. Raises a `ValueError` if there is no such item.

The optional arguments `start` and `end` are interpreted as in the slice notation and are used to limit the search to a particular subsequence of the list. The returned index is computed relative to the beginning of the full

RTFM

Definition (RTFM): The acronym
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RTFM

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RTFM

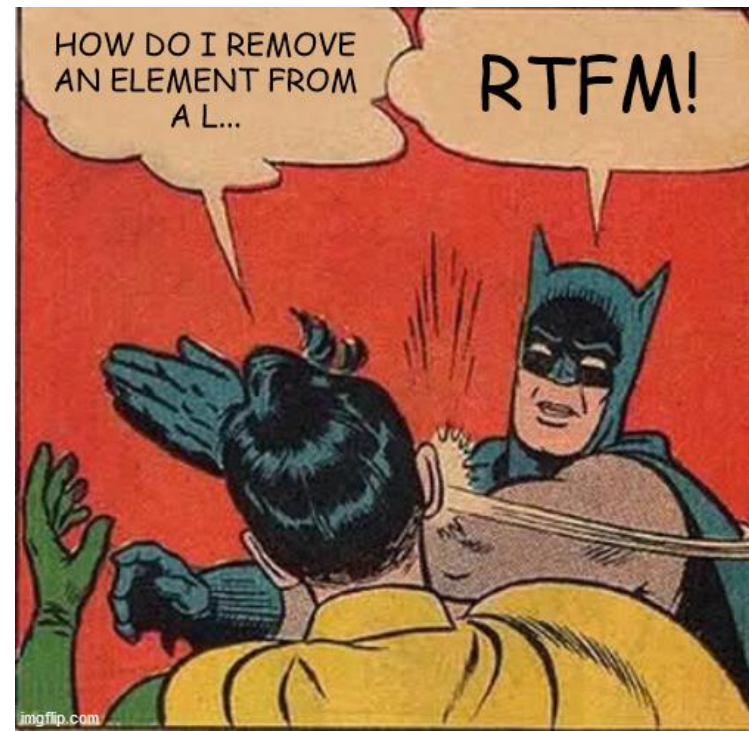
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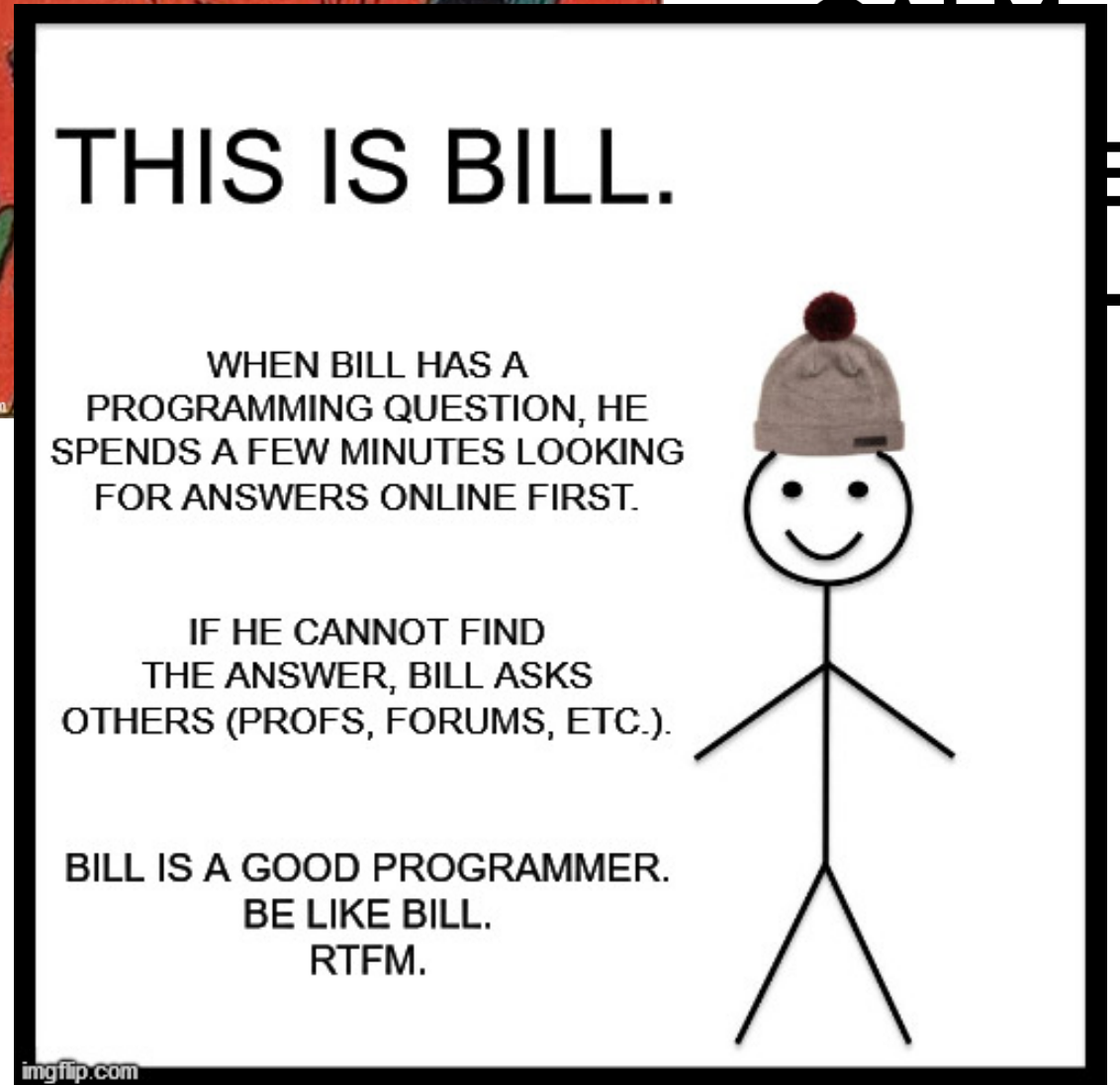
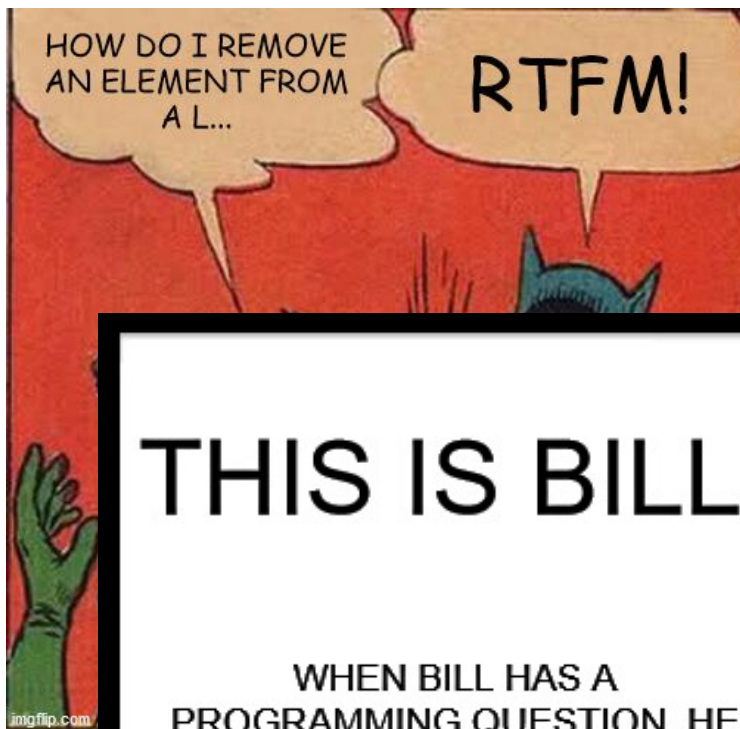

**KEEP
CALM
AND
READ THE
MANUAL**

RTFM

Definition (RTFM): The acronym **RTFM** stands for “**Read. The. F***ing. Manual.**”.

- While it may sound rude, **RTFM** is positively used in computer science communities, to encourage people to try to help themselves first, before seeking assistance from others (other being professors, classmates, StackOverflow, ChatGPT, etc.)

(<https://en.wikipedia.org/wiki/RTFM>)



Matt's Great advice

Matt's Great Advice: Do not fear the Python documentation.

No programmer in the world knows everything about every function, method and variable type in Python.

Instead, we know **where and how to find the information online**, when needed.

Make it a habit to look for the Python documentation when in doubt or looking for something specific!



Additional cool stuff on lists

If time allows...

“Pythonic” loops

Definition (**Pythonic loop**):

A Pythonic loop is a programming concept, where a list of elements is generated by integrating a **for** loop inside a list definition.

```
# Let us consider the list below  
my_list = [1, 3, 5, 7, 9, 11]  
print(my_list)
```

```
# We could create it by telling Python to use  
# a range and make a list out of it, like so.  
my_list = list(range(1, 13, 2))  
print(my_list)
```

```
# What if we wanted a list of squared values instead?  
my_list = [1, 4, 9, 16, 25, 36]  
print(my_list)
```

“Pythonic” loops

Definition (**Pythonic loop**):

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```
# We would have to do a for loop and progressively assemble all elements together  
# Start with an empty list  
my_list = []  
# For loop on all values to be squared (1-6)  
for value in range(1, 7):  
    # Squaring value  
    element = value**2  
    # Appending newly calculated element to the list  
    my_list.append(element)  
print(my_list)
```

```
[1, 4, 9, 16, 25, 36]
```

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for value in range(1, 7):  
    # Squaring value + Appending newly calculated element to the list  
    my_list.append(value**2)  
print(my_list)
```

```
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    my_list.append(value**2)  
print(my_list)
```

[1, 4, 9, 16, 25, 36]

```
# Pythonic Loop version of the list!  
my_list = [value**2 for value in range(1, 7)]  
print(my_list)
```


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print(my_list)
```

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[1, 4, 9, 16, 25, 36]
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# For loop on all values to be squared (1 to 6)  
for value in range(1, 7):  
    # Squaring value + Appending newly calculated value  
    my_list.append(value**2)  
print(my_list)
```

[1, 4, 9, 16, 25, 36]

```
# Pythonic Loop version of the previous code  
my_list = [value**2 for value in range(1, 7)]  
print(my_list)
```

[1, 4, 9, 16, 25, 36]

Why use **pythonic loops**?

- Convenient when the elements of the list can be defined regularly (e.g. all squared numbers from a to b , etc.)
- Less cumbersome than typing the elements, manually, especially for large lists.
- Also, looks damn cool.

Repetition: `*` operator overloading

Repetition: you can **repeat** a list ***n*** times (with ***n*** being an **int**), by multiplying a list with an int, using the `*` operator.

- The result of this repetition is a list where the elements are repeated ***n*** times.
- As if we add summed (`+`) the list with itself ***n*** times in a row.
- Similar to **multiplying** a **string** with an **int**.

```
1 a_list = [0, 1, 2]
2 a_number = 3
3 list_product = a_list*a_number
4 print(list_product)
```

```
[0, 1, 2, 0, 1, 2, 0, 1, 2]
```

```
1 a_string = "pika"
2 a_number = 3
3 some_stuff = a_string*a_number + "chu"
4 print(some_stuff)
```

```
pikapikapikachu
```

Element-wise removal

Element-wise removal: the **remove()** method removes the **element** passed as its **parameter**, from the list on which the method is applied.

- The **remove()** method changes the list as a result.
- **Note:** if the element appear multiple times, it is only removed once, when it is first encountered.

```
1 my_list = [1, 3, 5, 7, 9, 11]
2 print(my_list)
```

```
[1, 3, 5, 7, 9, 11]
```

```
1 # Element-wise removal
2 an_element = 7
3 my_list.remove(an_element)
4 print(my_list)
```

```
[1, 3, 5, 9, 11]
```

```
1 my_list = [1, 3, 5, 7, 9, 11, 7]
2 print(my_list)
```

```
[1, 3, 5, 7, 9, 11, 7]
```

```
1 # Element-wise removal
2 an_element = 7
3 my_list.remove(an_element)
4 print(my_list)
```

```
[1, 3, 5, 9, 11, 7]
```

Index-wise removal

Index-wise removal: you can remove an element at a given index by using the **del** function on the element itself.

- **Note:** the **del** function can be used on **any variable, of any type**, to remove it from the Python memory.

```
1 my_list = [1, 3, 5, 7, 9, 11, 7]
2 print(my_list)
```

```
[1, 3, 5, 7, 9, 11, 7]
```

```
1 # Index-wise removal
2 an_index = 4
3 del(my_list[an_index])
4 print(my_list)
```

```
[1, 3, 5, 7, 11, 7]
```

```
1 a_number = 10
2 print("First print: ", a_number)
3 del(a_number)
4 print("Second print: ", a_number)
```

```
First print: 10
```

```
-----
NameError                                Traceback (most recent call last)
<ipython-input-94-ee50ccff233f> in <module>
      2 print("First print: ", a_number)
      3 del(a_number)
----> 4 print("Second print: ", a_number)

NameError: name 'a_number' is not defined
```

Index-wise extraction

Index-wise extraction: you can **extract** the **element** of a **list**, at the **index** given as its **parameter**, by using the **pop()** method.

- As with other methods, it **changes the list** on which it is applied.
- The **output** of the **pop()** method consists of the **element**, which is extracted from the list, at the given index.

```
1 a_deck_of_cards = ["7 of Hearts", "8 of Diamonds", \
2                     "10 of Hearts", "6 of Clubs"]
3 print(a_deck_of_cards)
```

```
['7 of Hearts', '8 of Diamonds', '10 of Hearts', '6 of Clubs']
```

```
1 an_index = 2
2 # Index-wise Pop method on list
3 drawn_card = a_deck_of_cards.pop(an_index)
4 print(drawn_card)
5 print(a_deck_of_cards)
```

```
10 of Hearts
```

```
['7 of Hearts', '8 of Diamonds', '6 of Clubs']
```


The `index()` method

The `index()` method is also useful to check for membership and find an element in a list.

- It returns the **index** of the first occurrence of the element passed in parameter, in the list it applies to.
- Careful though, as it fails if no element in the list has the value passed as parameter.

```
1 my_list = [1, 7, 10, 7, 14]
2 print(my_list.index(7))
```

1

```
1 my_list = [1, 7, 10, 7, 14]
2 print(my_list.index(4))
```

ValueError

```
<ipython-input-4-2ed34508db49> in <mc
      1 my_list = [1, 7, 10, 7, 14]
----> 2 print(my_list.index(4))
```

ValueError: 4 is not in list

List alteration with the `extend()` method

Concatenation: A second list can be added to a first one with the `extend()` method. This method works almost like `append()`, as:

- It **changes the list** on which the `extend()` method is applied,
- And adds the elements of the second lists to the first one.
- At the end of the operation, **the first list will be changed**. No reassignment needed.

```
1 first_list = [0, 1, 2]
2 second_list = [1, 4, 7]
3 print(first_list)
4 print(second_list)
5 # Extend method on a list
6 first_list.extend(second_list)
7 print("-")
8 print(first_list)
9 print(second_list)
```

[0, 1, 2]

[1, 4, 7]

-

[0, 1, 2, 1, 4, 7]

[1, 4, 7]

Inserting an element in a list

Insertion: You can **insert** an **element**, at a given **index**, by using the **insert()** method.

- As with the **extend()** and **append()** methods, the **insert()** method is applied on a list.
- It receives an **index** as its **first parameter** and an **element** as its **second parameter**.
- And it **changes the list as a result**.

```
1 my_list = [1, 3, 5, 7, 9, 11]
2 print(my_list)
```

```
[1, 3, 5, 7, 9, 11]
```

```
1 an_element = 'Hello'
2 an_index = 4
3 my_list.insert(an_index, an_element)
4 print(my_list)
```

```
[1, 3, 5, 7, 'Hello', 9, 11]
```

Additional methods and functions on lists

Many other additional methods and functions can be used on lists.

- **Maximum:** the `max()` function returns the element with maximal value.
- **Minimum:** the `min()` function returns the element with maximal value.
- **Sum:** the `sum()` function returns the sum of elements in list.
- **Reverse:** the `reverse()` method reverses the elements in a list.

```
1 # Max and min values
2 grades_list = [85, 70, 80, 70, 65]
3 print("Max: ", max(grades_list))
4 print("Min: ", min(grades_list))
5 print("Sum: ", sum(grades_list))
```

```
Max: 85
Min: 65
Sum: 370
```

```
1 # Reverse
2 a_list = [7, 8, 4, 0, 1, 5]
3 a_list.reverse()
4 print(a_list)
```

```
[5, 1, 0, 4, 8, 7]
```

Additional methods and functions on lists

Many other additional methods and functions can be used on lists.

- **Sorting:** the `sort()` method can be used to sort the elements,
- In **ascending order** for **numerical** elements,
- And in **alphabetical order** for **string** elements.

Note: does not work on lists with mixed types elements.

```
1 # Sorting
2 a_list = [7, 8, 4, 0, 1, 5]
3 a_list.sort()
4 print(a_list)
```

```
[0, 1, 4, 5, 7, 8]
```

```
1 # Sorting
2 a_list = ["Matt", "Tony", "Oka", "Norman"]
3 a_list.sort()
4 print(a_list)
```

```
['Matt', 'Norman', 'Oka', 'Tony']
```

```
1 # Sorting
2 a_list = ["Matt", "Tony", 4, 7, 8]
3 a_list.sort()
4 print(a_list)
```

```
-----
TypeError                                Traceback (most recent call last)
<ipython-input-111-4fcabc28e0be2> in <module>
      1 # Sorting
      2 a_list = ["Matt", "Tony", 4, 7, 8]
----> 3 a_list.sort()
      4 print(a_list)
```

```
TypeError: '<' not supported between instances of 'int' and 'str'
```

Additional methods and functions on lists

Many other additional methods and functions can be used on lists.

- **Counting:** the `count()` method can be used to count how many times an element appears in a list.
- And many more things!

```
1 # Counting
2 a_list = ["Potion", "Sword", "Shield", "Potion", "Potion"]
3 number = a_list.count("Potion")
4 print(number)
```



Conclusion (Chapter 8)

- What is the **list type**, in detail, this time?
- What are the **indexing, slicing, updating** and **traversing** a list operations?
- What are **lists of lists** and their uses?
- How to **create, add** and **remove** elements from a list?
- What are advanced operations on list such as **concatenation, repetition, insertion**, etc?
- What does **RTFM** mean?
- **Practice** on lists.

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Which of these four lists (a,b,c and d) is empty, i.e. has no elements in it?

① Start presenting to display the poll results on this slide.

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What will be displayed?

① Start presenting to display the poll results on this slide.